

Dimitri Chrysafis

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Education

University of Wisconsin–Madison

Graduating May 2027

Bachelor of Science in Computer Science

GPA: 4.0

Relevant Coursework: COMP SCI 577 (Algorithms), COMP SCI 524 (Optimization), MATH 435 (Cryptography)

Technical Skills

ML Systems: PyTorch, CatBoost, LoRA/QLoRA fine-tuning, quantization, adapter merging

GPU/Graphics Systems: WebGPU/WGSL, WebGL/GLSL, Apple Metal, compute shaders, real-time rendering, mesh geometry, low-level GPU memory access

Experience

[SimpleX Chat](#)

Open Source Contributor Sep 2025–Present

- Improved multi-device synchronization in the Haskell client pipeline of an open-source messaging platform
- Implemented sync-state reconciliation and connection-lifecycle updates to reduce duplicate delivery edge cases across linked devices

Projects

Interactive 3D Fluid Simulator (MLS-MPM) [\[Demo\]](#) [\[GitHub\]](#)

- Built a browser-based 3D MLS-MPM simulator in JavaScript/WebGPU, keeping particle and grid state in GPU storage buffers and simulating **400,000 particles** with staged compute kernels
- Implemented fixed-point atomic accumulation for particle-to-grid transfers plus interactive dam-break, piston, particle-injection, and orbit-camera controls

GPU Fractal Rendering Engine [\[GitHub\]](#) [\[Blog\]](#)

- Built GPU fractal renderers in WebGL2/GLSL and Apple Metal for Newton, Kleinian, and Julia sets with interactive pan/zoom and shader-based iteration
- Implemented a Metal compute pipeline for **6,000×6,000** Julia frames with adaptive iteration counts and GPU-parallel image generation

GPU-Accelerated Sphere Packing in Arbitrary Meshes [\[GitHub\]](#) [\[Blog\]](#)

- Built a stochastic tangent-sphere packing solver for arbitrary OBJ meshes, enforcing interior, surface-clearance, and pairwise non-overlap constraints
- Accelerated ray-mesh containment, nearest-surface queries, and candidate search with batched PyTorch tensors on MPS/CUDA; visualized results in Open3D

Tennis Match Prediction System

- Trained a CatBoost ensemble on **92,000 ATP matches (1982–2024)**, engineering ELO, surface, head-to-head, recent-form, and schedule-fatigue features to achieve **83.7% accuracy**, outperforming the baseline by **12%**

Fourier Drawing Machine [\[GitHub\]](#) [\[Blog\]](#)

- Implemented a Discrete Fourier Transform pipeline with nearest-neighbor point ordering to reconstruct drawings as rotating epicycles
- Parallelized high-resolution frame rendering with Numba, PIL/NumPy, and a 14-worker executor for custom and image-derived point sets